Unit - 3

Jointly distributed RV

X(Head) = 1

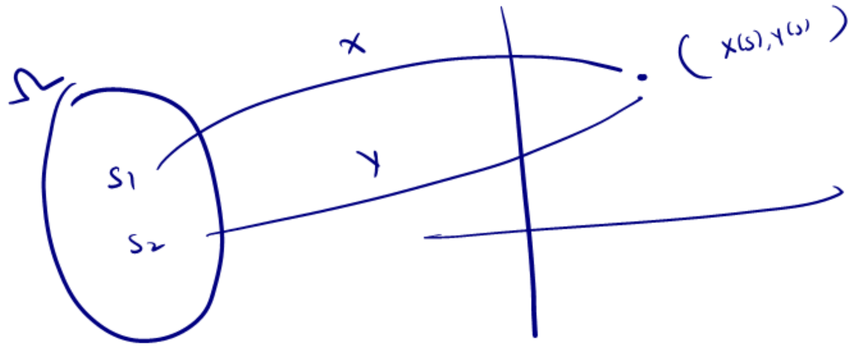
X → Univariate

Real world application → More one Variable

Pick a card of deck

(X, Y)

X Suite Y → denomination



→ Person

{X, Y, Z}

X – BP

Y – pulse

X → Univariate

X, Y ~ Bivariate

X, Y, Z → trivariate

X, Y, ... → Multivariate

X → Discrete → PMF → Bernoulli, Bin

→ CDF

X → Continuous → PDF →

CDF

(X, Y) ← Joint

Recall

Univariate

Discrete

① PMF $p_x(x_i) = P(x = x_i)$

$p_x(x) \geq 0$

$\sum_{\forall x_i} p_x(x_i) = 1$

Continuous

pdf $f(x)$

$\int_{-\infty}^{\infty} f(x)\,dx = 1$

$f(x) \geq 0$

$P(x \leq x) = \int_{-\infty}^{x} f(x)\,dx$

Bivariate

discrete

X, Y are 2 random variable

sample space

1. $p_{xy}(x_i, y_i) = P(x = x_i, y = y_i)$

2. $p_{x,y}(x_i, y_i) \geq 0$

3. $\sum_{\forall y}\sum_{\forall x_i} p_{xy}(x_i, y_i) = 1$

continuous

Continuous

$f_{xy}(x,y)$

$P(a < x < b, c < y < d)$

$= \int_c^d \int_a^b f_{xy}(x,y)\,dx\,dy$

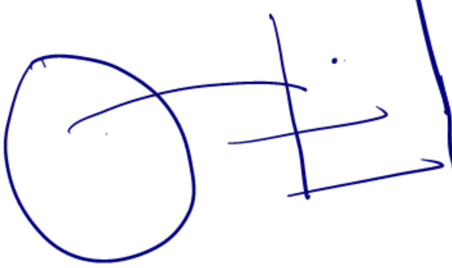
$= \int_a^b \int_c^d f_{xy}(x,y)\,dy\,dx$

$f_{xy}(x,y) \geq 0$

$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{xy}(x,y)\,dx\,dx = 1$

X, Y → k

Single



→ $p_{xy}(x,y)$ → Joint PMF

$f_{xy}(x,y)$ → Joint pdf

Univariate

X	0	1	2
$p_x(x_i)$			

Ref: Bivariate Probability distribution table

$X \backslash Y$	y_1	y_2	$\cdots$	y_m	
x_1	p_{11}	p_{12}	$\cdots$	p_{1m}	$p_X(x_1)$ — marginal dis of $p_X(x_1)$
x_2	p_{21}	p_{22}	$\cdots$	p_{2m}	$p_X(x_2)$
$\vdots$					
x_n	p_{n1}	p_{n2}	$\cdots$	p_{nm}	$p_X(x_n)$
	$p_Y(y_1)$	$p_Y(y_2)$		$p_Y(y_m)$	1

$$\sum_{\forall y}\sum_{\forall x} p_{XY}(x,y) =$$

→ Marginal distribution (discrete)

→ Let X, Y be 2 jointly distributed r.v. with joint PMF $p_{XY}(x_i, y_j)$

→ The individual distribution of X, Y are called marginal distribution.

→ Marginal distribution of function X

$$p_X(x) = \sum p_{XY}(x, y_j) = \sum_{\forall y_j} p(x, y_j) = \sum_{y} p(x,y)$$

— Marginal dist of function Y

$$p_Y(y) = \sum_{\forall x_i} p_{XY}(x_i, y)$$

→ Suppose a car showroom has 10 cars of a brand out of which 5 are good (G), 2 are of defective transmission (DT), 3 are of defective steering (DS). Now if 2 cars are selected at random. Find joint pmf.

$X \sim$ denote number of cars with DT

$Y \sim$ denote number of cars with DS

10 → 5 Good, 2 DT, 3 DS

$$P(0,0) = P(X=0, Y=0) = \frac{{}^5C_2}{{}^{10}C_2} = \frac{2}{9}$$

$$P(X=0, Y=1) = \frac{{}^5C_1 \times {}^3C_1}{{}^{10}C_2}$$

X \ Y (DS)	0	1	2	
0	2/9	1/3	1/15	$p_X(0)$
1 (DT)	2/9	2/15	0	$p_X(1)$
2	1/45	0	0	$p_X(2)$
	$p_Y(0)$	$p_Y(1)$	$p_Y(2)$	1

Marginal dist of X

${}^2C_2$

$$P(X=1, Y=0) = \frac{{}^2C_1 \times {}^5C_1}{{}^{10}C_2} = \frac{2}{9}$$

$$P(X=2, Y=2)$$

$$P(X=1, Y=1) = \frac{{}^2C_1 \times {}^3C_1}{{}^{10}C_2} =$$

$$P(X \le 1, Y \le 1) = P(0,0) + P(0,1) + P(1,0) + P(1,1)$$

$$= \frac{41}{45}$$